



1062CH07

COORDINATE GEOMETRY

7

7.1 Introduction

In Class IX, you have studied that to locate the position of a point on a plane, we require a pair of coordinate axes. The distance of a point from the y -axis is called its **x -coordinate**, or **abscissa**. The distance of a point from the x -axis is called its **y -coordinate**, or **ordinate**. The coordinates of a point on the x -axis are of the form $(x, 0)$, and of a point on the y -axis are of the form $(0, y)$.

Here is a play for you. Draw a set of a pair of perpendicular axes on a graph paper. Now plot the following points and join them as directed: Join the point A(4, 8) to B(3, 9) to C(3, 8) to D(1, 6) to E(1, 5) to F(3, 3) to G(6, 3) to H(8, 5) to I(8, 6) to J(6, 8) to K(6, 9) to L(5, 8) to A. Then join the points P(3.5, 7), Q(3, 6) and R(4, 6) to form a triangle. Also join the points X(5.5, 7), Y(5, 6) and Z(6, 6) to form a triangle. Now join S(4, 5), T(4.5, 4) and U(5, 5) to form a triangle. Lastly join S to the points (0, 5) and (0, 6) and join U to the points (9, 5) and (9, 6). What picture have you got?

Also, you have seen that a linear equation in two variables of the form $ax + by + c = 0$, (a, b are not simultaneously zero), when represented graphically, gives a straight line. Further, in Chapter 2, you have seen the graph of $y = ax^2 + bx + c$ ($a \neq 0$), is a parabola. In fact, coordinate geometry has been developed as an algebraic tool for studying geometry of figures. It helps us to study geometry using algebra, and understand algebra with the help of geometry. Because of this, coordinate geometry is widely applied in various fields such as physics, engineering, navigation, seismology and art!

In this chapter, you will learn how to find the distance between the two points whose coordinates are given. You will also study how to find the coordinates of the point which divides a line segment joining two given points in a given ratio.

7.2 Distance Formula

Let us consider the following situation:

A town B is located 36 km east and 15 km north of the town A. How would you find the distance from town A to town B without actually measuring it. Let us see. This situation can be represented graphically as shown in Fig. 7.1. You may use the Pythagoras Theorem to calculate this distance.

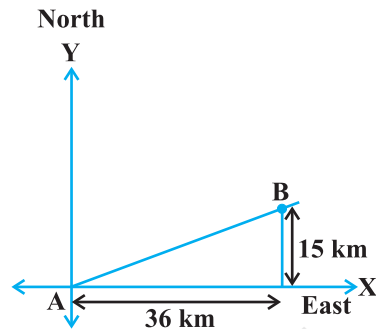


Fig. 7.1

Now, suppose two points lie on the x-axis. Can we find the distance between them? For instance, consider two points A(4, 0) and B(6, 0) in Fig. 7.2. The points A and B lie on the x-axis.

From the figure you can see that $OA = 4$ units and $OB = 6$ units.

Therefore, the distance of B from A, i.e., $AB = OB - OA = 6 - 4 = 2$ units.

So, if two points lie on the x-axis, we can easily find the distance between them.

Now, suppose we take two points lying on the y-axis. Can you find the distance between them. If the points C(0, 3) and D(0, 8) lie on the y-axis, similarly we find that $CD = 8 - 3 = 5$ units (see Fig. 7.2).

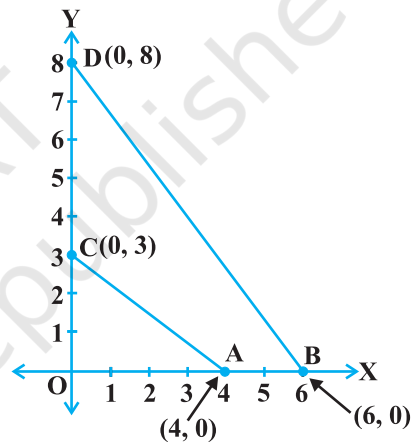


Fig. 7.2

Next, can you find the distance of A from C (in Fig. 7.2)? Since $OA = 4$ units and $OC = 3$ units, the distance of A from C, i.e., $AC = \sqrt{3^2 + 4^2} = 5$ units. Similarly, you can find the distance of B from D = $BD = 10$ units.

Now, if we consider two points not lying on coordinate axis, can we find the distance between them? Yes! We shall use Pythagoras theorem to do so. Let us see an example.

In Fig. 7.3, the points P(4, 6) and Q(6, 8) lie in the first quadrant. How do we use Pythagoras theorem to find the distance between them? Let us draw PR and QS perpendicular to the x-axis from P and Q respectively. Also, draw a perpendicular from P on QS to meet QS at T. Then the coordinates of R and S are (4, 0) and (6, 0), respectively. So, $RS = 2$ units. Also, $QS = 8$ units and $TS = PR = 6$ units.

Therefore, $QT = 2$ units and $PT = RS = 2$ units.

Now, using the Pythagoras theorem, we have

$$\begin{aligned} PQ^2 &= PT^2 + QT^2 \\ &= 2^2 + 2^2 = 8 \end{aligned}$$

So, $PQ = 2\sqrt{2}$ units

How will we find the distance between two points in two different quadrants?

Consider the points $P(6, 4)$ and $Q(-5, -3)$ (see Fig. 7.4). Draw QS perpendicular to the x -axis. Also draw a perpendicular PT from the point P on QS (extended) to meet y -axis at the point R .

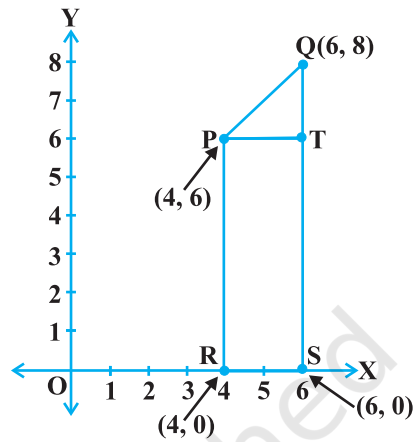


Fig. 7.3

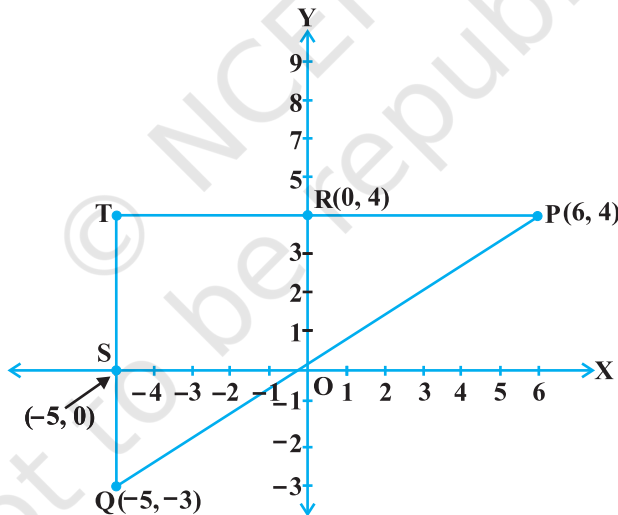


Fig. 7.4

Then $PT = 11$ units and $QT = 7$ units. (Why?)

Using the Pythagoras Theorem to the right triangle PTQ , we get

$$PQ = \sqrt{11^2 + 7^2} = \sqrt{170} \text{ units.}$$

9. If $Q(0, 1)$ is equidistant from $P(5, -3)$ and $R(x, 6)$, find the values of x . Also find the distances QR and PR .
10. Find a relation between x and y such that the point (x, y) is equidistant from the point $(3, 6)$ and $(-3, 4)$.

7.3 Section Formula

Let us recall the situation in Section 7.2. Suppose a telephone company wants to position a relay tower at P between A and B is such a way that the distance of the tower from B is twice its distance from A . If P lies on AB , it will divide AB in the ratio $1 : 2$ (see Fig. 7.9). If we take A as the origin O , and 1 km as one unit on both the axis, the coordinates of B will be $(36, 15)$. In order to know the position of the tower, we must know the coordinates of P . How do we find these coordinates?

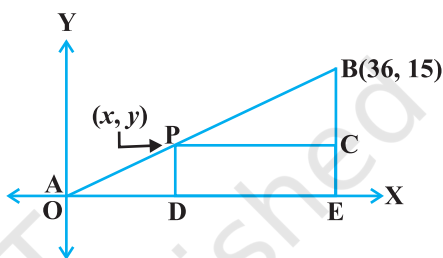


Fig. 7.9

Let the coordinates of P be (x, y) . Draw perpendiculars from P and B to the x -axis, meeting it in D and E , respectively. Draw PC perpendicular to BE . Then, by the AA similarity criterion, studied in Chapter 6, $\triangle POD$ and $\triangle BPC$ are similar.

Therefore, $\frac{OD}{PC} = \frac{OP}{PB} = \frac{1}{2}$, and $\frac{PD}{BC} = \frac{OP}{PB} = \frac{1}{2}$

So, $\frac{x}{36-x} = \frac{1}{2}$ and $\frac{y}{15-y} = \frac{1}{2}$.

These equations give $x = 12$ and $y = 5$.

You can check that $P(12, 5)$ meets the condition that $OP : PB = 1 : 2$.

Now let us use the understanding that you may have developed through this example to obtain the general formula.

Consider any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ and assume that $P(x, y)$ divides AB internally in the ratio $m_1 : m_2$, i.e.,

$$\frac{PA}{PB} = \frac{m_1}{m_2} \quad (\text{see Fig. 7.10}).$$

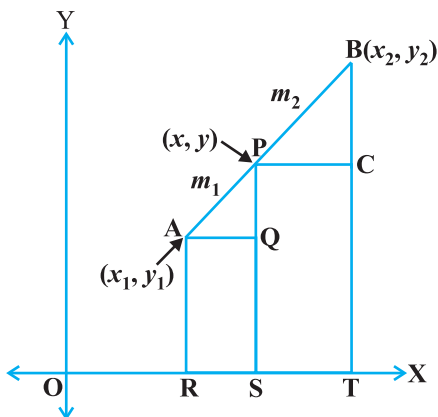


Fig. 7.10

Draw AR, PS and BT perpendicular to the x-axis. Draw AQ and PC parallel to the x-axis. Then, by the AA similarity criterion,

$$\Delta PAQ \sim \Delta BPC$$

Therefore,
$$\frac{PA}{BP} = \frac{AQ}{PC} = \frac{PQ}{BC} \quad (1)$$

Now,
$$\begin{aligned} AQ &= RS = OS - OR = x - x_1 \\ PC &= ST = OT - OS = x_2 - x \\ PQ &= PS - QS = PS - AR = y - y_1 \\ BC &= BT - CT = BT - PS = y_2 - y \end{aligned}$$

Substituting these values in (1), we get

$$\frac{m_1}{m_2} = \frac{x - x_1}{x_2 - x} = \frac{y - y_1}{y_2 - y}$$

Taking
$$\frac{m_1}{m_2} = \frac{x - x_1}{x_2 - x}, \text{ we get } x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

Similarly, taking
$$\frac{m_1}{m_2} = \frac{y - y_1}{y_2 - y}, \text{ we get } y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

So, the coordinates of the point P(x, y) which divides the line segment joining the points A(x₁, y₁) and B(x₂, y₂), internally, in the ratio m₁ : m₂ are

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) \quad (2)$$

This is known as the **section formula**.

This can also be derived by drawing perpendiculars from A, P and B on the y-axis and proceeding as above.

If the ratio in which P divides AB is k : 1, then the coordinates of the point P will be

$$\left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1} \right).$$

Special Case : The mid-point of a line segment divides the line segment in the ratio 1 : 1. Therefore, the coordinates of the mid-point P of the join of the points A(x₁, y₁) and B(x₂, y₂) is

$$\left(\frac{1 \cdot x_1 + 1 \cdot x_2}{1 + 1}, \frac{1 \cdot y_1 + 1 \cdot y_2}{1 + 1} \right) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Let us solve a few examples based on the section formula.

7.4 Summary

In this chapter, you have studied the following points :

1. The distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
2. The distance of a point $P(x, y)$ from the origin is $\sqrt{x^2 + y^2}$.
3. The coordinates of the point $P(x, y)$ which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m_1 : m_2$ are $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$.
4. The mid-point of the line segment joining the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$.

A NOTE TO THE READER

Section 7.3 discusses the Section Formula for the coordinates (x, y) of a point P which divides internally the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m_1 : m_2$ as follows :

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \quad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$$

Note that, here, $PA : PB = m_1 : m_2$.

However, if P does not lie between A and B but lies on the line AB , outside the line segment AB , and $PA : PB = m_1 : m_2$, we say that P divides externally the line segment joining the points A and B . You will study Section Formula for such case in higher classes.